



Amazon Pharmacy Improves Customer Care Using Amazon Bedrock and Amazon SageMaker

Learn how Amazon Pharmacy reduces administrative burden in healthcare by leveraging Amazon SageMaker.

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99% of prescriptions

include up-front pricing estimates

Architected

to promote HIPAA compliance for customer privacy

Optimized

operational efficiency



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Overview

With growing demand in the pharmacy industry, pharmacists face an increasingly heavy administrative burden that takes the focus away from customers. In an environment where inefficient legacy systems still dominate, Amazon Pharmacy is using Amazon Bedrock and Amazon SageMaker to improve customer care.



Pharmacy is working to improve the customer experience with the help of artificial intelligence (AI).

Founded in 2020, [Amazon Pharmacy](#) is a full-service digital pharmacy on Amazon.com that is available in all 50 US states. Using HIPAA-eligible AI and machine learning (ML) services on Amazon Web Services (AWS), Amazon Pharmacy is improving the customer experience by delivering prescription medications faster, providing up-front insurance estimates, and offering ongoing access to clinical and customer care.



Opportunity | Helping Customers Access Medications Quickly

Studies consistently show that [20–30 percent of Americans](#) never fill their prescriptions. “Because of affordability and access issues, people often don’t take medications that are important for their longitudinal health,” says John Love, vice president of Amazon Pharmacy. “Our goal is to help more people access the medications that support longer, healthier lives.”

Since its launch, Amazon Pharmacy developed AI and ML solutions on AWS, beginning with demand forecasting to improve medication delivery times. Then, Amazon Pharmacy saw opportunities where generative AI could bring greater immediate value to customers, which included providing estimated insurance pricing, improving data entry, and assisting the customer care team to find information more quickly.

Solution | Using Generative AI and ML to Offer Price Transparency and Enhance Customer Support

Using HIPAA-eligible generative AI and ML tools on AWS, Amazon Pharmacy is providing its customers with greater price transparency. For example, in 2023, Amazon Pharmacy began offering insurance pricing estimates for 99 percent of prescriptions, using deep learning statistical models hosted on [Amazon SageMaker](#)—a service where data scientists and ML engineers can build, train, and deploy ML models for virtually any use case with fully managed infrastructure, tools, and workflows. The price estimates help customers to make more informed decisions. Once a customer's prescription is sent to Amazon Pharmacy, the customer sees clear, transparent, personalized insurance and cash pay options, discounts available through PrimeRx, and other potential ways to save before checkout. "These are huge wins for the customers," says Love. "They are getting more transparent pricing while receiving their medications quickly."

In 2023, Amazon Pharmacy began looking at how chatbots based on large language models could help it enhance customer support. Using Amazon SageMaker, Amazon Pharmacy created an [LLM-based chatbot](#) so that its customer care representatives could focus on improving the customer experience. The chatbot helps to save significant amounts of time for Amazon Pharmacy's customer care agents. "If you go into any brick-and-mortar pharmacy, you're at the whim of one individual getting every detail right, among all the things they're trying to do," says Love. "Through our use of AI, there are no distractions."

Amazon Pharmacy used two models to create its chatbot: an embedding model and a large language model. The first model helps with indexing retrieval and is crucial in extracting relevant answers from Amazon Pharmacy's extensive help center knowledge base. The development team experimented quickly with different models using [Amazon SageMaker JumpStart](#)—an ML hub with foundation models, built-in algorithms, and prebuilt ML solutions that can be deployed with just a few clicks. Using Amazon SageMaker JumpStart, the Amazon Pharmacy development team cut months of work that they otherwise would have needed to train models from scratch.

The second model summarizes the extracted information for the customer care representatives to review, relying on foundation models accessed through [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models from major AI companies along with a broad set of capabilities that organizations need to build generative AI applications with security, privacy, and responsible AI. Customer care representatives ask questions and receive responses using natural language, which helps boost productivity. "When customers call our support center with questions, our models pull up information at our representatives' fingertips," says Love. "The speed and quality of this process improves customer access."

Outcome | Building Efficiencies into the Customer Experience

Amazon Pharmacy uses additional AWS services to help make the customer experience seamless. To transcribe prescriptions that doctors' offices send by fax, Amazon Pharmacy uses [Amazon Textract](#)—a service that automatically extracts printed text, handwriting, and data from virtually any document with intelligent document processing. "On AWS, we've achieved combinatorial innovation by bringing together pieces of technology to solve issues," says Love. "We can flexibly use different technologies to increasingly improve healthcare."

Amazon Pharmacy structures data for prescription processing using its own custom solution and [Comprehend Medical](#), a HIPAA-eligible, natural-language-processing service that uses ML that  ned to understand and extract health data from unstructured medical text.

“We use generative AI in our customer support to extract payer information, think about next-action recommendations, vet insurance, and help us interpret all the data we receive,” says Alex Alves, senior principal engineer at Amazon Pharmacy. “It makes the process faster and more accurate. Using AWS, we can customize solutions for our industry while prioritizing security and privacy.”

About Amazon Pharmacy

Amazon Pharmacy is a full-service pharmacy on Amazon.com. It offers transparent pricing, clinical and customer support, and free delivery right to customers' doors.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

Amazon Textract

Amazon Textract is a machine learning (ML) service that automatically extracts text, handwriting, layout elements, and data from scanned documents.

[Learn more »](#)

Amazon Comprehend Medical

Amazon Comprehend Medical is a HIPAA-eligible natural language processing (NLP) service that uses machine learning that has been pre-trained to understand and extract health data from medical text, such as prescriptions, procedures, or diagnoses.

[Learn more »](#)



Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

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